DAY 10 – LINUX FILES, DIRECTORIES & PERMISSIONS

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THEORY – ANSWERS (ALL 20 QUESTIONS)

1. What is a Linux directory?

A Linux directory is a folder used to store and organize files and other directories.

2. What does mkdir do?

mkdir is used to create a new directory (folder) in Linux.

3. What does touch do?

touch is used to create an empty file if it does not exist. It can also update the file's timestamp.

4. What does cat do?

cat is used to display the contents of a text file in the terminal.

5. What does cp do?

cp is used to copy files from one location to another without removing the original file.

6. What does mv do?

mv is used to move files or directories from one location to another. It can also be used to rename files.

7. What does rm do?

rm is used to remove or delete files in Linux.

8. What does ls -l display?

ls -l displays files and directories in a detailed list, including file permissions, owner, group, file size, date and name.

9. Explain r, w and x permissions.

Linux uses three main file permissions:

- **r (Read) = 4:** Allows a user to read or view the contents of a file.
- **w (Write) = 2:** Allows a user to write, edit or modify a file.
- **x (Execute) = 1:** Allows a user to execute a file or script.
- **- (No Permission) = 0:** Indicates that the permission is not allowed.

Linux permissions are assigned to three categories:

1. **User (u) / Owner:** The user who owns the file.
2. **Group (g):** A group of users assigned to the file.
3. **Others (o):** All users who are neither the owner nor members of the assigned group.

Permission Numeric Values:

Permission	Symbol	Numeric Value
Read	r	4
Write	w	2
Execute	x	1
No Permission	-	0

Example: -rwxr-xr--

- First - = File type
- rwx = User/Owner (4+2+1 = 7)
- r-x = Group (4+0+1 = 5)
- r-- = Others (4+0+0 = 4)

10. Explain user, group and others.

Linux permissions are divided into three categories:

- **User (u) / Owner:** The person who owns the file.
- **Group (g):** Users belonging to the file's assigned group.
- **Others (o):** All other users who are neither the owner nor part of the group.

11. What does chmod do?

chmod stands for Change Mode. It is used to change the permissions of files and directories in Linux.

12. What does chmod 755 represent?

chmod 755 gives the owner read, write and execute permissions, while the group and others get read and execute permissions.

- **User/Owner (7):** Read (4) + Write (2) + Execute (1) = rwx
- **Group (5):** Read (4) + Execute (1) = r-x

- **Others (5):** Read (4) + Execute (1) = r-x

Therefore, chmod 755 represents rwxr-xr-x.

13. What does chmod 644 represent?

chmod 644 gives the owner read and write permissions, while the group and others get only read permission.

- **User/Owner (6):** Read (4) + Write (2) = rw-
- **Group (4):** Read (4) = r--
- **Others (4):** Read (4) = r--

Therefore, chmod 644 represents rw-r--r--.

14. What is file ownership?

File ownership identifies the user and group associated with a file. It determines who owns the file and helps manage its permissions.

15. What is the purpose of /etc?

/etc is a directory that stores system configuration files used to manage Linux settings and services.

16. What is the purpose of /home?

/home is a directory that contains the personal directories of regular users. It stores their personal files and data.

17. What is commonly stored under /var?

/var stores variable data that changes during system operation, such as system logs and other changing application data.

18. Why should rm be used carefully?

rm should be used carefully because it permanently deletes files and Linux does not provide a recycle bin by default. Deleting the wrong file can cause data loss.

19. Why are Linux permissions important?

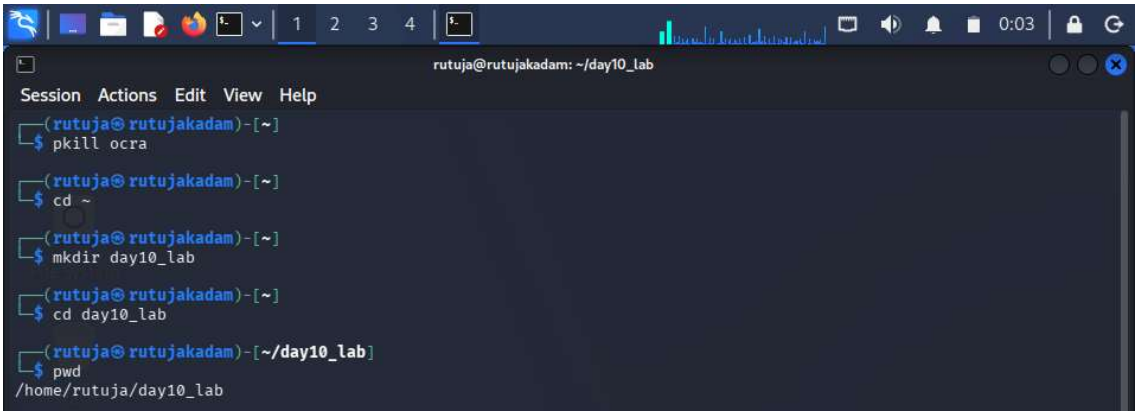
Linux permissions are important because they protect files and directories from unauthorized access, modification and execution. They help maintain system security.

20. How can incorrect permissions create a security risk?

Incorrect permissions can allow unauthorized users to read sensitive files, modify important data or execute harmful scripts. This can lead to data theft, data loss and security problems.

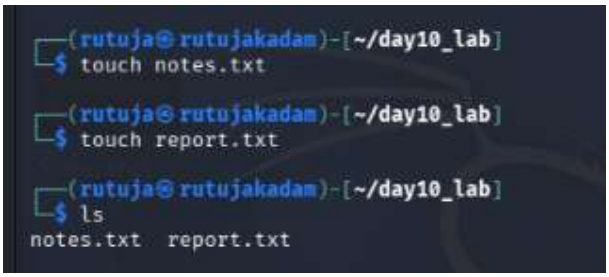
DAY 10 PRACTICAL LAB — Linux File Management & Permissions Lab

- **SETUP:** cd ~ · mkdir day10_lab · cd day10_lab · pwd (verify you are inside the lab directory)



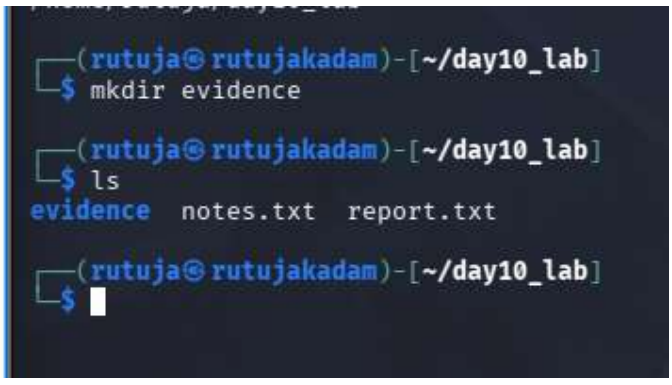
TASK 1 —

- Create files: touch notes.txt · touch report.txt
- ls → you should see both files.

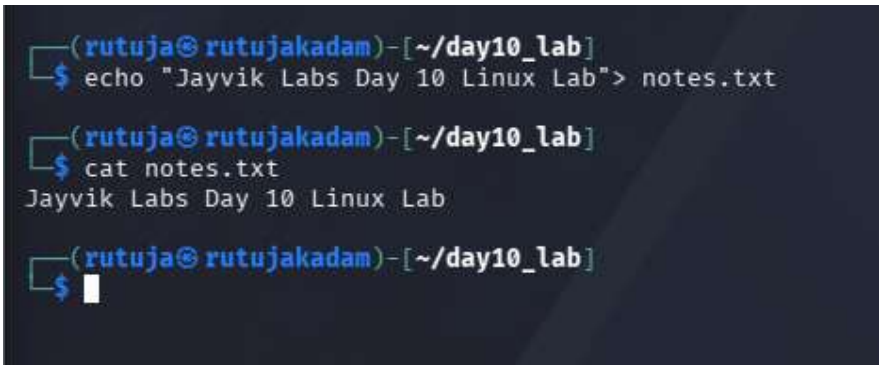


TASK 2 —

- Create directory: mkdir evidence
- ls.

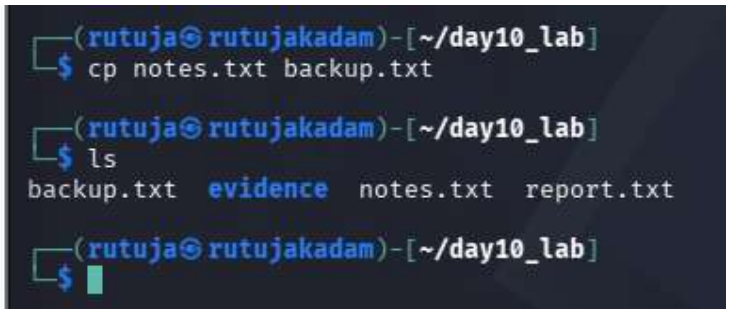


TASK 3 — Add content: echo "Jayvik Labs Day 10 Linux Lab" > notes.txt → cat notes.txt. The > writes command output into a file (lab files only).



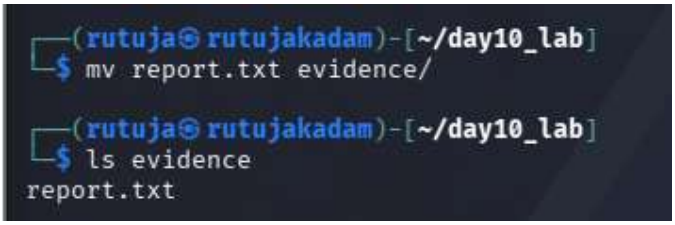
TASK 4 —

- Copy: cp notes.txt backup.txt
- ls → backup.txt exists.



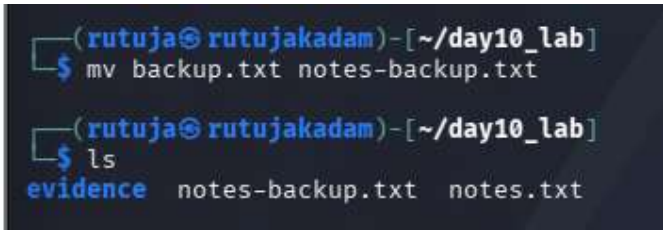
TASK 5 —

- Move: mv report.txt evidence/
- ls evidence → report.txt moved in.



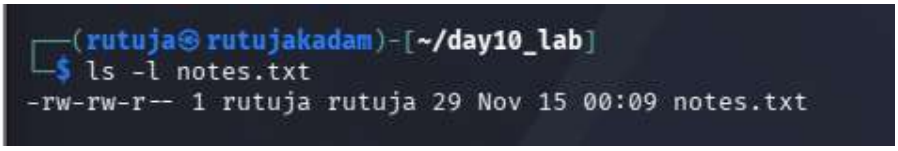
TASK 6 —

- Rename: mv backup
- txt notes-backup.txt · ls.



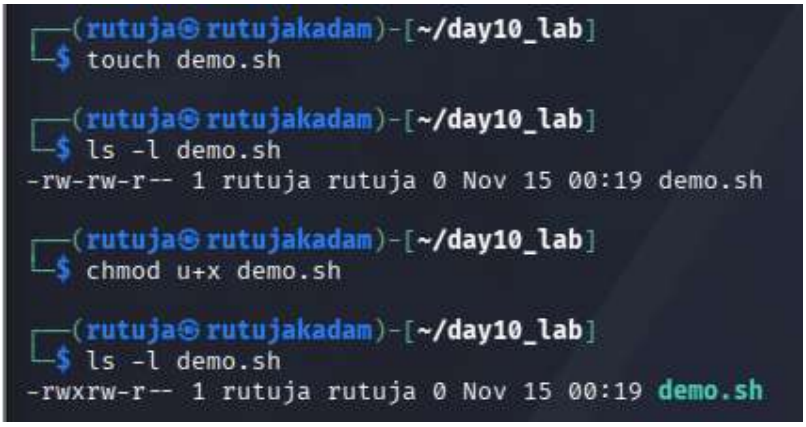
TASK 7 —

- View permissions: ls -l notes.txt → identify Owner / Group / Others permissions.



TASK 8 —

- Change permissions: touch demo.sh
- ls -l demo.sh · chmod u+x demo.sh
- ls -l demo.sh → what changed?



TASK 9 —

- Numeric permissions: chmod 644 notes.txt
- ls -l notes.txt → explain Owner r/w, Group r, Others r.

```
┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ chmod 644 notes.txt

┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ ls -l notes.txt
-rw-r--r-- 1 rutuja rutuja 29 Nov 15 00:09 notes.txt
```

Meaning:

- User/Owner (6) = Read (4) + Write (2)
- Group (4) = Read only
- Others (4) = Read only

TASK 10 —

- Safe delete: touch delete-me.txt
- ls · rm delete-me.txt
- ls. LAB FILE ONLY — always verify the name before rm.

```
┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ touch delete-me.txt

┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ ls
delete-me.txt  demo.sh  evidence  notes-backup.txt  notes.txt

┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ rm delete-me.txt

┌──(rutuja㉿rutujakadam)-[~/day10_lab]
└─$ ls
demo.sh  evidence  notes-backup.txt  notes.txt
```